

5. Project Development Phase — Preprocessing & Business Questions

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Objective

This phase prepares the four raw EV datasets in MySQL Workbench and maps specific analytical business questions, framed around the project's three personas (Aarav, Meera, Ravi), to the Tableau worksheets and dashboards built from that data.

Data Preprocessing Steps

- **EVIndia (12 rows):** Row count check, duplicate-car check, and TRIM on Car, Style, Transmission, and VehicleType. No duplicate rows were found; row count remained 12 after cleaning.
- **ElectricCarData_Clean (103 → 98 rows):** Checked for duplicate Brand/Model combinations (5 found and removed), executed TRIM on Brand, Model, RapidCharge, PowerTrain, and BodyStyle, and validated that Range_Km, TopSpeed_KmH, and PriceEuro were all positive values.
- **Charging Station Data (202 → 168 rows):** TRIM on region, address, aux addres, type, power, and service fields; validated latitude/longitude fell within valid coordinate bounds; built a DISTINCT clean table (charging_station_clean, 168 rows) to remove exact duplicate station entries.
- **Cheapest EV Database (180 rows):** Checked for missing PriceinGermany and PriceinUK values (several rows have one or the other listed as "N/A", which was preserved rather than dropped, since each row still has at least one usable price field), executed TRIM on Name, Subtitle, and Drive, and checked for duplicate car names. No duplicate names were found; row count remained 180.

Business Questions Map

Each business question below is mapped to the actual Tableau worksheet(s) built to answer it.

#	Business Question	Tableau Worksheet(s)
1	How does EV range vary by market segment?	Average Range by Segment
2	Which brands and powertrains are represented in the dataset?	Brand filtered by PowerTrain type (+ pie/treemap variants)
3	Which EVs provide the strongest driving range relative to retail price?	Range vs Price; Cost per km of Range
4	Which manufacturers maintain the highest fast-charge ratios?	Fast Charge Ratio by Brand
5	Which vehicles achieve optimal electricity efficiency?	Range vs Efficiency Scatter; Top 10 most efficient EV Brands
6	What is the relationship between top speed and acceleration?	Top Speed vs Acceleration
7	Which Indian EVs are available, and how do prices/ranges compare?	Different EV cars in India; Price for different cars in India; India Price vs R
8	How does charging station density vary by region?	Charger Count by Region; Charging Stations by region and type in India*
9	Where are public charging stations located?	Map of Charger Locations; EV Charging stations map of India*
10	How does charger power capacity (kW) vary by region?	Power Capacity by Region

#	Business Question	Tableau Worksheet(s)
11	What is the breakdown of charging station service types?	Service Type Breakdown**

* **Note:** the worksheets "Charging Stations by region and type in India" and "EV Charging stations map of India" are named using "India", but the underlying charging station dataset covers Delhi only. This naming should be corrected to "Delhi" for accuracy before final submission.

** **Note:** every station in the cleaned charging dataset is recorded as "Self Service" — there is no attended-service category present in the data, so this breakdown currently shows a single category rather than a genuine split.